
★ Collecting Primary Data

We collected our data using the NC State University's Centennial Campus using the Driving Simulator to test autonomous behaviors in a controlled environment. Data was processed and analyzed via RStudio to evaluate system performance across all 39 configurations.

★ Staying Under Budget

Leveraging university-provided tools and hardware, the project was completed with a total cost of **\$0**, excluding the cost of printing and the trifold.

★ Attended the Driving Simulator

The FEDD team visited the driving simulator to gain a deeper understanding of its technical operations and functions for this project. Although technical difficulties prevented us from actually operating the simulator ourselves, the visit allowed us to receive a comprehensive, in-depth walkthrough of the system, which allowed us to learn exactly how the trials worked.

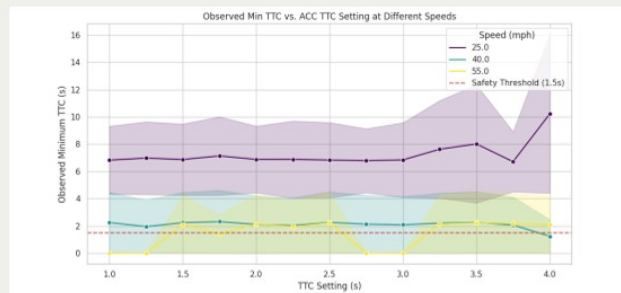
★ Submitted the Research Component on Time

Our team submitted our research component to our class TA a day before scheudle and submitted our final research documentation before FEDD day.

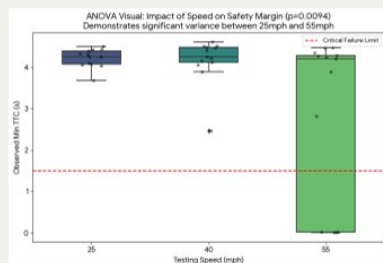
SAFETY

★ minimizes collision risk based on simulation results and adherence to safe distances

According to industry-standard safety models, a 1.50-second headway is the mandatory boundary needed to sum human reaction time and brake lag; any observed value falling below this red line is a system failure.



The graph illustrates that while standard settings (1.50s) function perfectly at low speeds, high velocity (55 mph, shown by the yellow line) can cause severe failures, with several trials dropping to zero. Our final recommendation of a 2.00s TTC baseline provides a crucial 33% Factor of Safety.



This research proved that the ACC logic must include consider speed which can be seen in our final reccomendations, as a safe urban setting becomes dangerous at 55 mph without a built-in safety buffer.

DRIVING THE FUTURE OPTIMIZING ADAPTIVE CRUISE CONTROL FOR AUTONOMOUS VEHICLES

E101(018) Team 4, 2026

Meet the Team

Narayani Garg

Manager, Point of Contact, Data Analysis through R programming

- My role was to conduct the comprehensive data analysis and design the optimized parameter sets based on the findings.

Milan Borovicnin

Modeler, Scheduler

- My role was to establish project timelines and map out the step-by-step logic of how Adaptive Cruise Control (ACC) functions.

Breanna Mexico

Researcher

- My role was to find the crucial background information and build the foundational research before starting the project.

Ian Richards

Designer, Quality Assurance

- My role was to design the process visuals and conduct verifications against the deliverable requirements.